QPCODE

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION MONTH 2025
THIRD SEMESTER
DSC-B-MG3DSCBIF202-Sequence Analysis Using Bioinformatics
(2024 ADMISSION ONWARDS)

Duration: 1.5hrs **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Describe PAM. [U] / [CO1]

2. Describe HMM. [U] / [CO2]

3. Analyse the use of MEME. [An] / [CO3]

4. Discover the use of AUGUSTUS tool in gene prediction. [A] / [CO4]

5.	Examine the use of Genbank.	[A] / [CO5]
6.	Outline the tools for Multiple Sequence Alignment.	[An] / [CO3]
7.	Compare identity, similarity and homology.	[U] / [CO1]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

8.	Summarize Needleman Wunch algorithm.	[U] / [CO1]
9.	Differentiate patterns, profile and motif.	[U] / [CO2]
10.	Classify MSA tools.	[An] / [CO3]
11.	Examine the role of Genemark.	[A] / [CO4]
12.	Distinguish between PSI BLAST and PHI- BLAST.	[U] / [CO2]
13.	Describe Multiple Sequence Alignment.	[U] / [CO1]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **2** questions
Each question carries **10** marks

14.	Explain MSA tools.	[U] / [CO1]
15.	Interpret gene prediction and annotation.	[A] / [CO5]
16.	Examine Gene Prediction tools.	[An] / [CO3]
17.	Summarize Sequence Patterns and Profiles.	[U] / [CO2]

[10x2 = 20]